Ayemere J. Bellowalker

Bioinformatics Scientist

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SUMMARY

Bioinformatics Scientist with 3+ years of experience developing scalable, reproducible genomics workflows for large-scale sequencing datasets (>2.3 billion reads; 190GB+). Proven expertise in workflow orchestration (Nextflow), containerisation (Docker), and cloud-based computing (AWS EC2).

Specialised in metagenomics, microbiome analysis, and multi-omics integration, with a strong focus on building production-ready pipelines for automated data processing, genome reconstruction, and functional annotation. Passionate about developing robust bioinformatics systems to support large-scale biological data resources and collaborative research infrastructure.

TECHNICAL SKILLS

Programming & Workflow Development:
Python, R, Bash, Nextflow, Git

Bioinformatics & Genomics:
Metagenomics, 16S rRNA analysis, MAG reconstruction, microbiome analysis

Tools & Databases:
QIIME2, Kraken2, MEGAHIT, MetaBAT2, SemiBin2, GTDB-Tk, eggNOG, Mothur

Infrastructure & Compute:
AWS EC2, Docker, Linux/Unix, HPC

Data Science:
Statistical modelling, clustering, data visualisation

PROFESSIONAL EXPERIENCE

Doctoral Researcher
October 2022 - Present

Harper Adams University - Telford, UK

- Designed and deployed scalable bioinformatics pipelines for metagenomic and amplicon datasets (>2.3 billion reads), reducing manual processing time by ~30%
- Built modular, reproducible workflows using Nextflow and Docker, enabling portable execution across environments
- Executed large-scale analyses on AWS EC2 infrastructure, optimising compute efficiency for high-throughput datasets
- Developed end-to-end metagenomic workflows integrating taxonomic classification (Kraken2), assembly (MEGAHIT), binning (MetaBAT2, SemiBin2), and annotation (GTDB-Tk, eggNOG)
- Reconstructed 1,429 metagenome-assembled genomes (MAGs) and implemented automated QC pipelines to generate 255 medium/high-quality genomes (≥50–90% completeness; ≤5–10% contamination)
- Identified 80 putative novel microbial genomes and detected 469 nitrogen metabolism genes via functional annotation workflows
- Integrated multi-omics datasets for predictive modelling of microbial systems and metabolic function
- Produced well-documented, version-controlled workflows supporting reproducible research and collaborative development

Laboratory Technician
July 2022 - September 2022

Intertek - Stoke-on-Trent, UK

- Performed high-throughput PCR and ELISA assays on 200+ food samples daily, ensuring accurate and timely data generation in a regulated laboratory environment.
- Contributed to microbial and chemical risk assessments through precise sample preparation, assay execution, and ISO-compliant data recording.
- Maintained strict adherence to biosafety, contamination control, and quality management protocols within a GLP-aligned workflow.

PROJECTS

Scalable Metagenomic Analysis Pipeline (Nextflow + AWS)

GitHub Repository: Scalable-Metagenomic-Analysis-Pipeline

Developed a modular, reproducible metagenomics pipeline using Nextflow DSL2 integrating Kraken2, MEGAHIT, MetaBAT2, CheckM2, and GTDB-Tk for taxonomic classification, genome assembly, genome binning, and MAG quality assessment.

- Processed large-scale sequencing datasets (>2.3 billion reads; 190GB)
- Designed cloud-compatible workflows using AWS and containerised environments
- Generated high-quality MAGs and functional annotations for nitrogen metabolism
- Implemented automated reporting using MultiQC

Amplicon Analysis Pipeline (QIIME2 + R)

GitHub Repository: Amplicon-Analysis-Pipeline

- Developed a complete 16S rRNA analysis workflow using QIIME2, including denoising (DADA2), taxonomic classification (SILVA), and diversity analysis
- Performed statistical analysis using R (phyloseq, DESeq2, vegan), including differential abundance testing and multivariate modelling
- Implemented reproducible workflows for microbiome community analysis and visualisation

EDUCATION

Doctor of Philosophy (Ph.D.)
October 2022 - Present

Harper Adams University - Telford, UK

- Research focus: Enhancing the role of the rumen microbiome in improving nitrogen use efficiency in dairy cows
- Specialisation in genomics, bioinformatics, computational biology, microbiology, molecular biology, cell biology, and nutrition

Bachelor of Science (Honours)
September 2019 - July 2022

Staffordshire University - Stoke-on-Trent, UK

1st Class Honours

- Coursework included Cancer Biology & Regenerative Medicines, Genomics & Bioinformatics, Current Advances and Bioethics, Microbial genomes & pathogenesis

CERTIFICATIONS

- Attended & presented at 14th International Gut Microbiome Symposium, France (June 2025)
- Linux for Genomics Workshop – Edinburgh Genomics, University of Edinburgh (March 2025)